

Lab: Bayes' Theorem -- Updating Beliefs with Evidence

Objective

Apply **Bayes' theorem** to update beliefs when new evidence arrives. This is the core mathematical operation of cognition in Active Inference: combining what you already believe (prior) with what you observe (likelihood) to form a new belief (posterior).

Prerequisites

- Completed Math Foundations: Perception module (conditional probability, likelihood)
- Comfort with fraction multiplication and division

Part 1: Bayes' Theorem by Counting

Goal: Derive Bayes' theorem from a concrete counting scenario.

A class of 100 students: 30 study regularly, 70 do not. Of those who study, 90% pass the exam. Of those who do not study, 40% pass.

1. Fill in the table:

	Pass	Fail	Total
Study	?	?	30
No Study	?	?	70
Total	?	?	100

1. $P(\text{Study} \mid \text{Pass}) = (\text{students who study AND pass}) / (\text{total who pass}) = ?$
2. Now compute the same answer using Bayes' formula: $P(\text{Study} \mid \text{Pass}) = P(\text{Pass} \mid \text{Study}) * P(\text{Study}) / P(\text{Pass}) = ?$

3. Verify both methods give the same answer.

Write your response here

Part 2: Bayes' Theorem Formula

Goal: Practice the formula $P(H|D) = P(D|H) * P(H) / P(D)$.

A disease affects 1 in 1000 people. A test is 99% accurate (detects disease when present) and has a 5% false positive rate.

1. $P(\text{disease}) = 0.001$, $P(\text{no disease}) = 0.999$
2. $P(\text{positive} | \text{disease}) = 0.99$
3. $P(\text{positive} | \text{no disease}) = 0.05$
4. $P(\text{positive}) = P(\text{pos}|\text{disease})P(\text{disease}) + P(\text{pos}|\text{no disease})P(\text{no disease}) = ?$
5. $P(\text{disease} | \text{positive}) = P(\text{pos}|\text{disease}) * P(\text{disease}) / P(\text{positive}) = ?$
6. Were you surprised by how low $P(\text{disease} | \text{positive})$ is? Why does this happen?

Write your response here

Part 3: Sequential Updating

Goal: Apply Bayes' theorem multiple times as new evidence arrives.

You think a coin might be fair ($P(\text{heads}) = 0.5$) or biased ($P(\text{heads}) = 0.8$). Your prior: $P(\text{fair}) = 0.7$, $P(\text{biased}) = 0.3$.

1. You flip and get heads. Compute $P(\text{fair} | \text{heads})$ using Bayes' theorem.

2. Now use your answer from step 1 as the new prior. You flip again and get heads. Compute $P(\text{fair} \mid \text{heads, heads})$.
3. One more flip: tails. Compute the new posterior.
4. How did your belief shift across three updates? In Active Inference, this sequential updating IS cognition.

Write your response here

Part 4: Prior Strength

Goal: Explore how strong vs. weak priors affect belief updating.

Use the same coin scenario, but compare two agents: - Agent A (strong prior): $P(\text{fair}) = 0.95$ - Agent B (weak prior): $P(\text{fair}) = 0.50$

Both observe 3 heads in a row.

1. Compute $P(\text{fair} \mid 3 \text{ heads})$ for Agent A.
2. Compute $P(\text{fair} \mid 3 \text{ heads})$ for Agent B.
3. Which agent shifts their belief more? Why?
4. In Active Inference, prior strength relates to **precision** -- how confident the agent is in its existing model.

Write your response here

Summary Table

Math Concept	Symbol	Definition
Bayes' theorem	$P(H D) = P(D H) * P(H) / P(D)$	Updates prior belief with evidence
Prior	$P(H)$	Belief before observing data
Posterior	$P(H D)$	Belief after observing data
Evidence / marginal likelihood	$P(D)$	Total probability of the data
Sequential updating	Prior -> Posterior -> new Prior	Repeated application of Bayes' rule